Visual Perception & Cognition

Designing for the mind behind the eyes

Xianbin Gu · Lecture 2

You can see everything—but what did you actually notice?

Visualization works through a human “processor”

Data → Visual representation → Perception → Cognition → Understanding (insight / decision)

A design succeeds only when people can **notice**, **organize**, and **interpret** what matters.

Today: from seeing to designing

01	**Seeing is active**	Attention selects; knowledge guides interpretation.
02	**The mind organizes**	Gestalt cues create groups, paths, and objects.
03	**Principles become decisions**	Notice, group, follow, and interpret.

Part 01

Seeing is active

The visual system filters and organizes before conscious reasoning begins.

Your eyes are not a camera

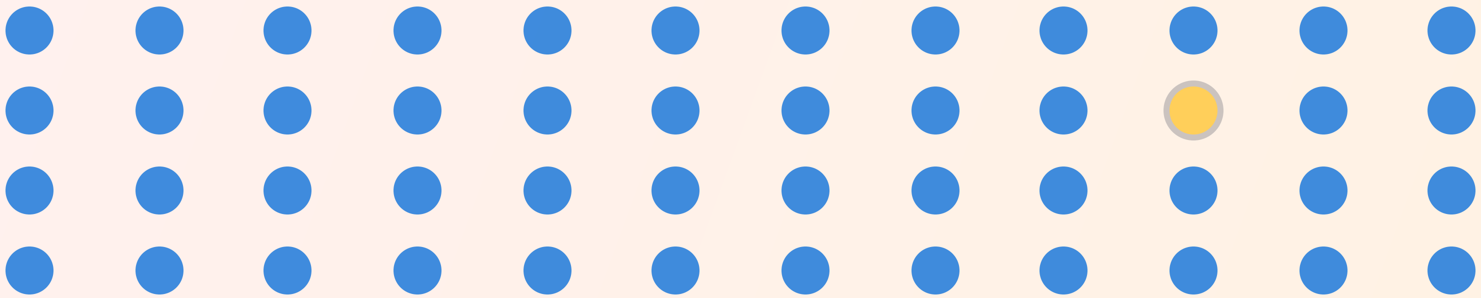
Light reaches the retina (visual stimulus) → Features and patterns are extracted (perception) → Attention selects → Patterns are recognized (cognition)

Edges, contrast, color, orientation, and motion are processed—not copied as an untouched picture.

Perception tells colors, shapes, and their positions in the figure; attention tells what you should care about; cognition recognizes them by comparing them with what you've known.

QUICK SCAN · 5 SECONDS

Find the one item that is different



Some visual differences guide attention automatically

Bottom-up

Driven by the stimulus

- contrast
- color difference
- orientation
- motion

Top-down

Driven by the viewer

- goal
- expectation
- prior knowledge
- current task

The same scene supports different searches

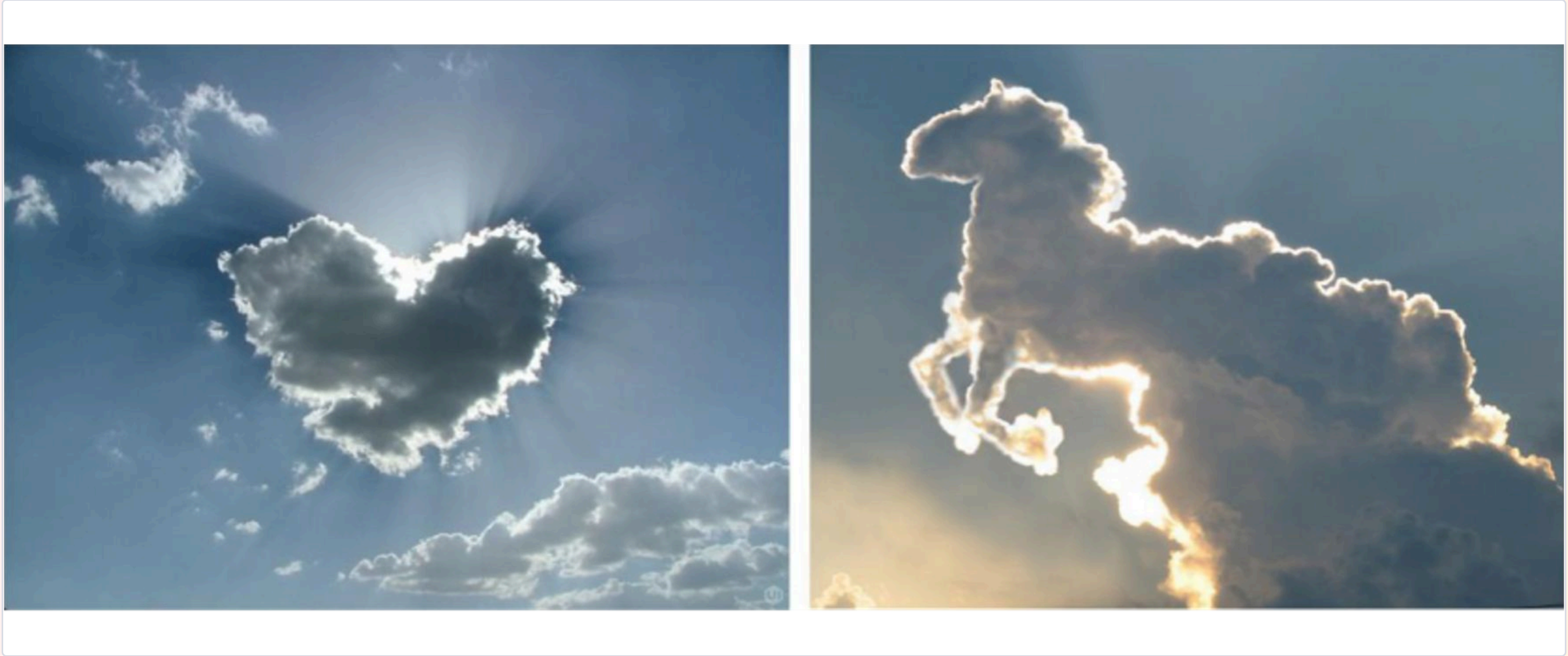
Your task changes what you see

Search for the person.

Now, search for the equations.

The stimulus is unchanged; the goal redirects attention.

What animals can you find?



Individual scan · 30 seconds · Then compare with a partner

Prior knowledge can dominate interpretation

What do you see first?

Once one interpretation is suggested, it becomes difficult to “unsee.” Perception and cognition continually constrain one another.

Perception and cognition form a loop

Visual evidence features & patterns	⇄	**Attention** what is selected now	⇄	**Knowledge & goals** what the viewer expects

Good visualization supports the loop; misleading visualization exploits it.

Perception and cognition form a loop

Good visualization supports the loop

The shape of the steering wheel reminds you the brand of the car.

Perception and cognition form a loop

Misleading visualization exploits it

William Ely Hill, My wife and my mother-in-law, 1915

WHY VISUALIZATION HELPS

> Visualization is about external cognition: using resources outside the mind to increase what the mind can do.

Paraphrased from Stuart K. Card

- External resources become useful evidence only when the visual system organizes separate marks into meaningful groups, paths, and figures.
- Gestalt principles explain how that organization happens.

Part 02

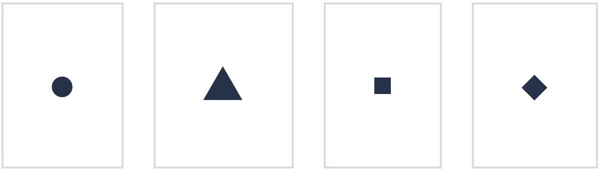
The mind organizes

Gestalt principles explain why separate marks become groups, paths, and objects.

We perceive organized wholes—not isolated marks

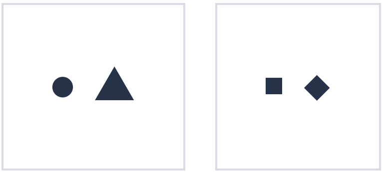
Without structure

Every item competes as an independent object.

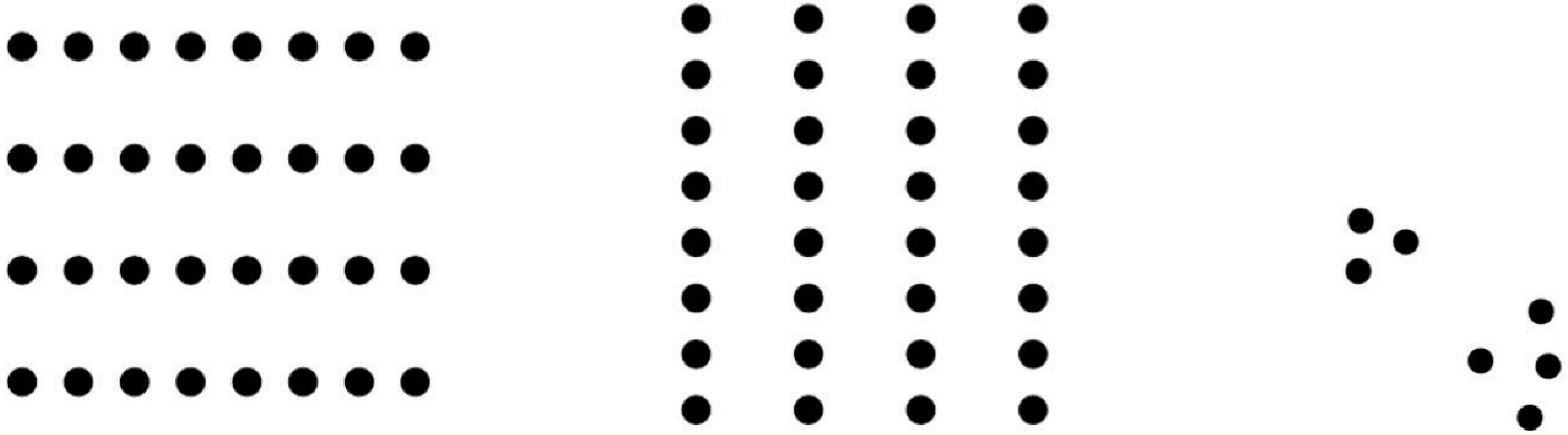


With structure

Relations let the mind compress many items into a few groups.

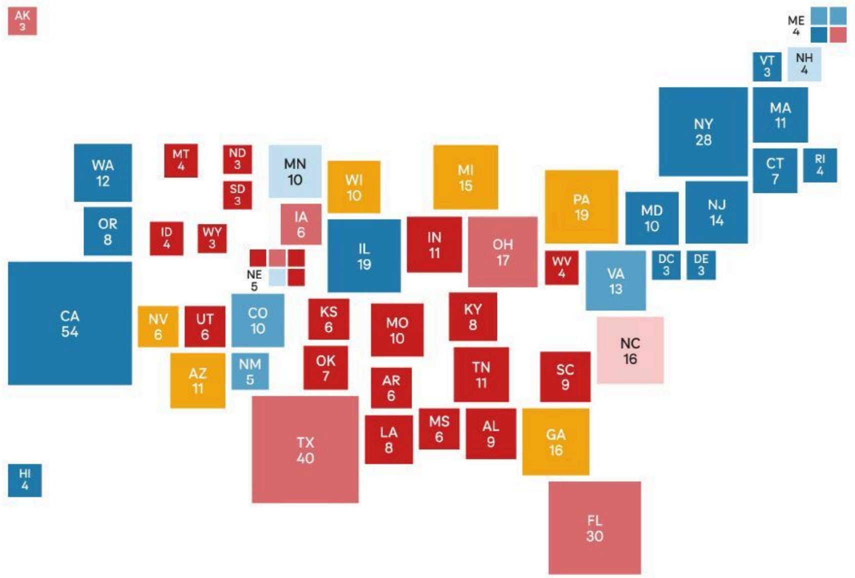


Proximity makes groups before labels do



Use whitespace deliberately: distance is not empty—it encodes relationship.

Using proximity in design

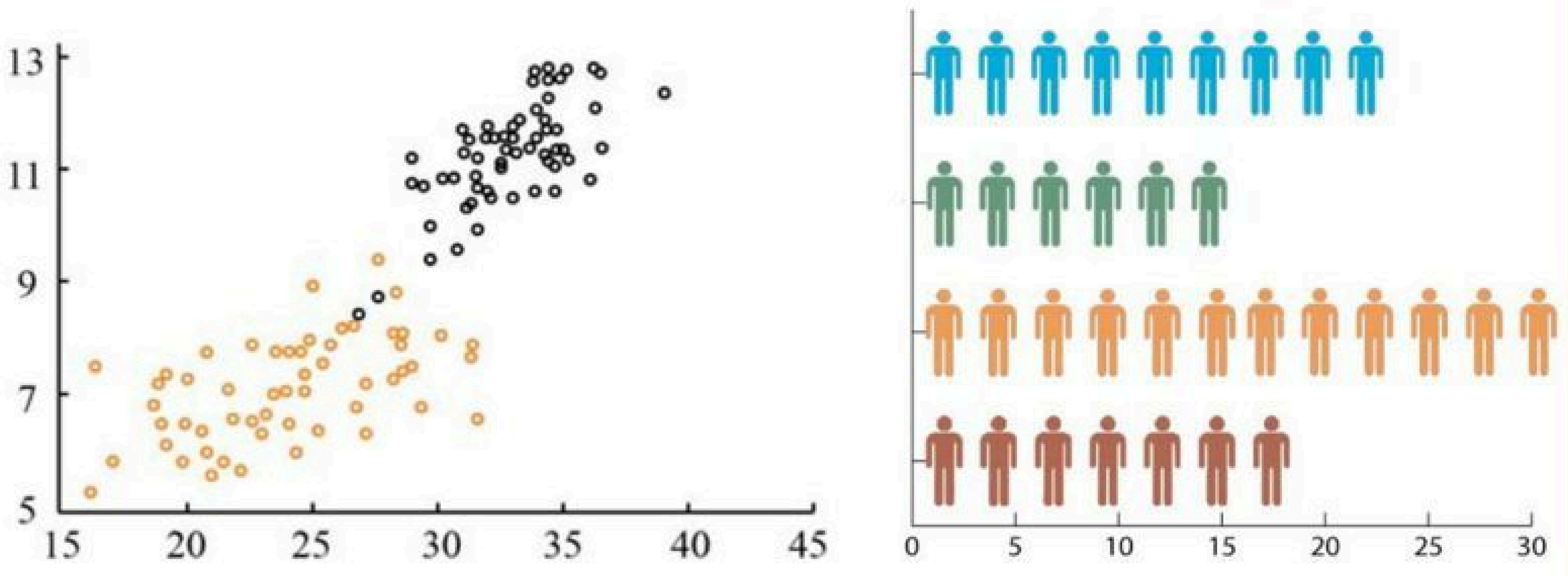


Nearby tiles form a recognizable map

Nearby symbols form a unified logo

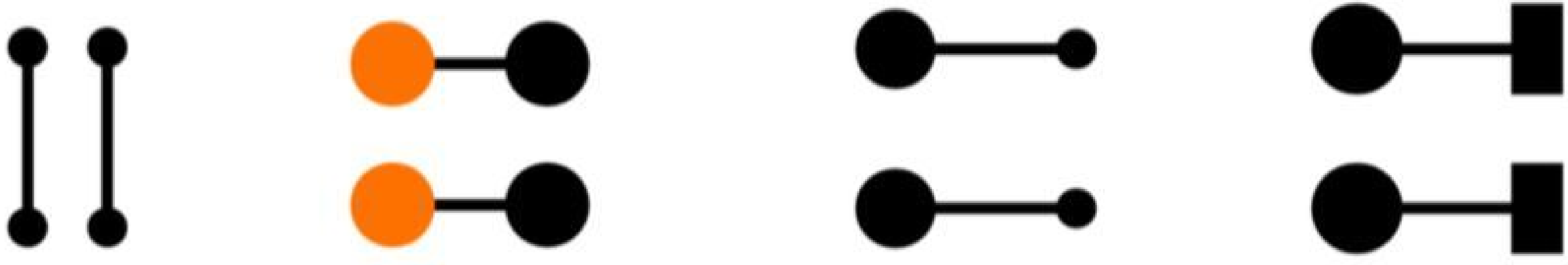
Proximity creates structure. Spacing helps viewers perceive separate elements as meaningful groups and larger forms.

Similarity says “these belong together”



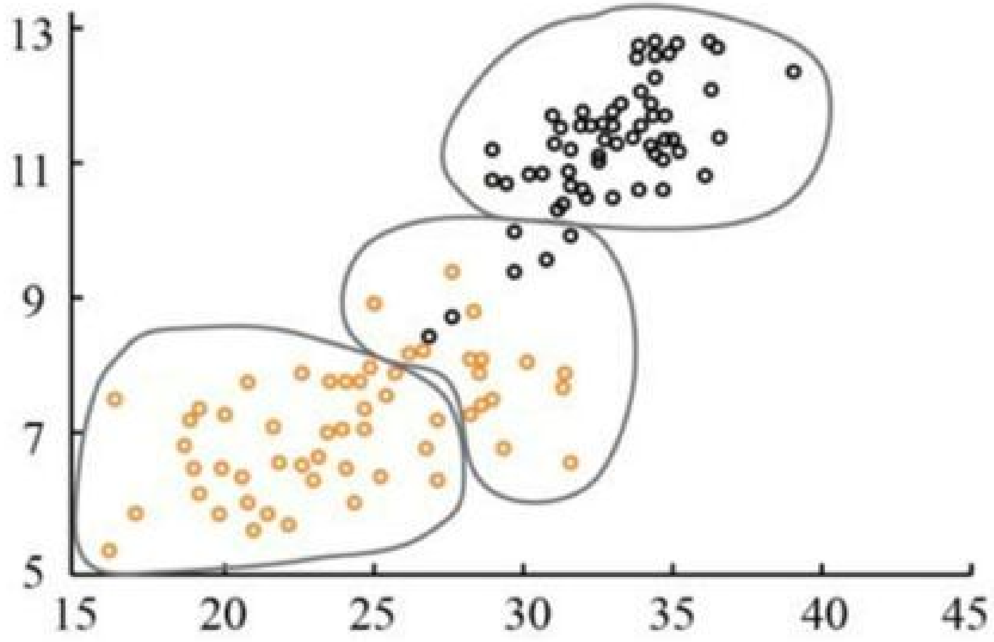
Color, shape, size, and orientation can create categories—even when none were intended.

Connectedness can override proximity and similarity



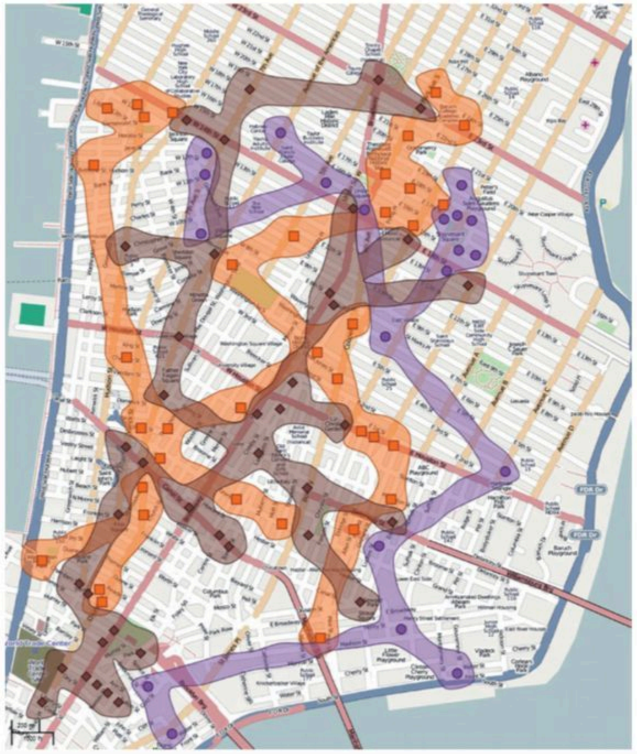
Use links for explicit relationships—but every line is a strong claim.

Enclosure turns boundaries into groups



Enclose related elements

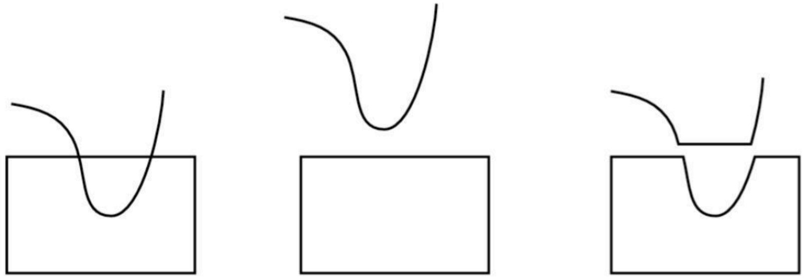
Elements within the same boundary are perceived as a group.



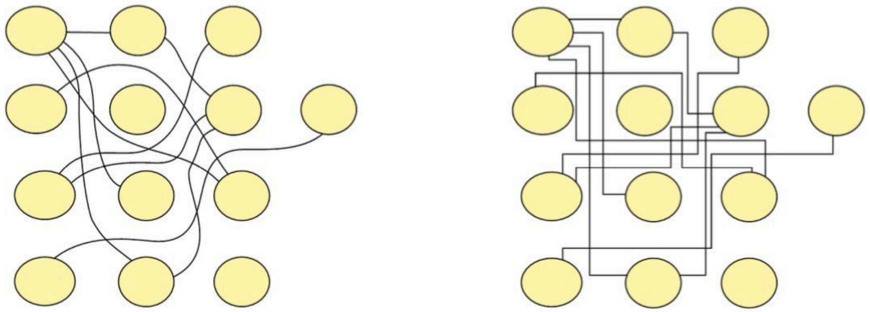
Use regions to reveal groups

Contours can make spatial distributions readable.

Continuity helps the eye follow a path

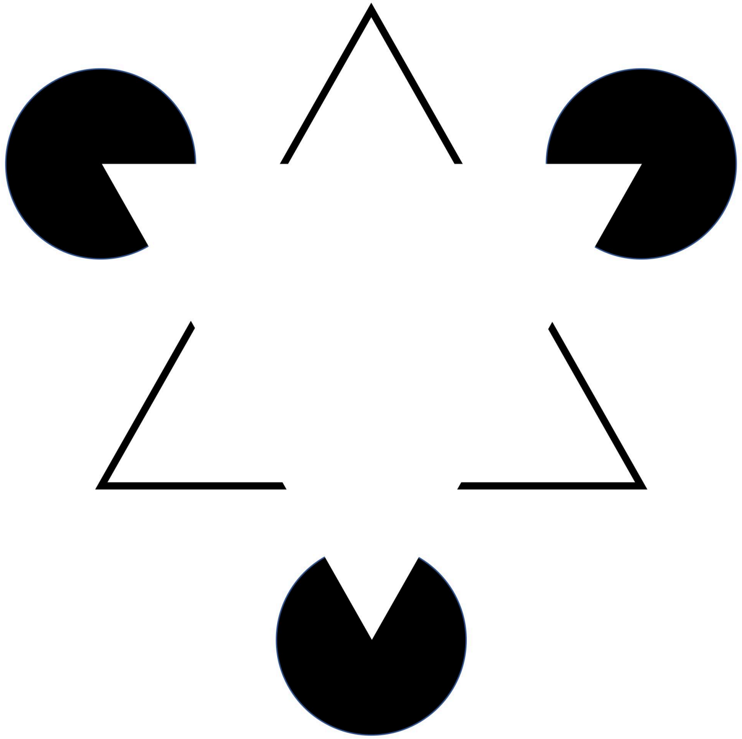


Continuous boundaries are perceived as a whole.

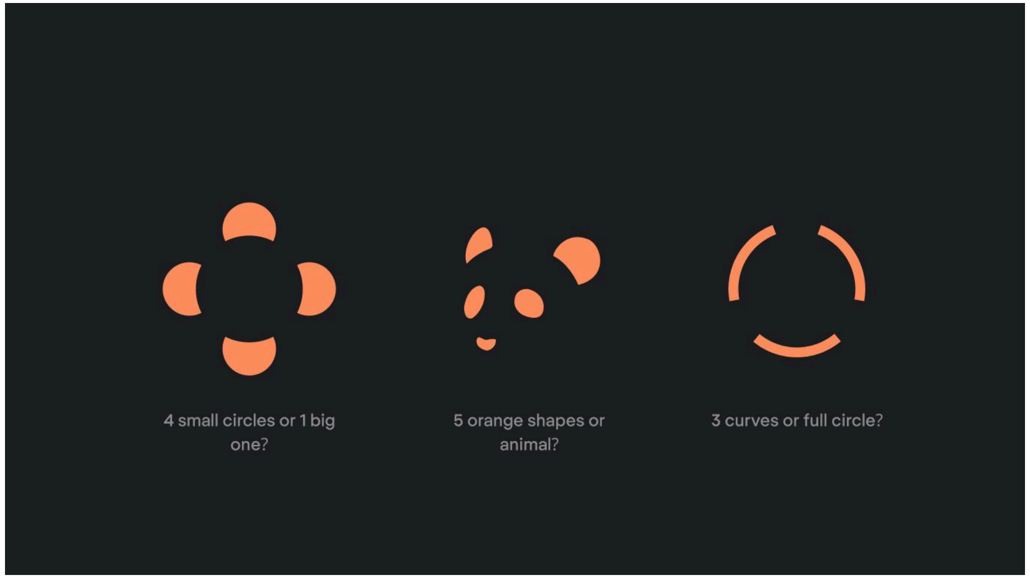


Smooth paths are easier to follow. Abrupt turns make paths harder to trace.

Closure completes incomplete objects

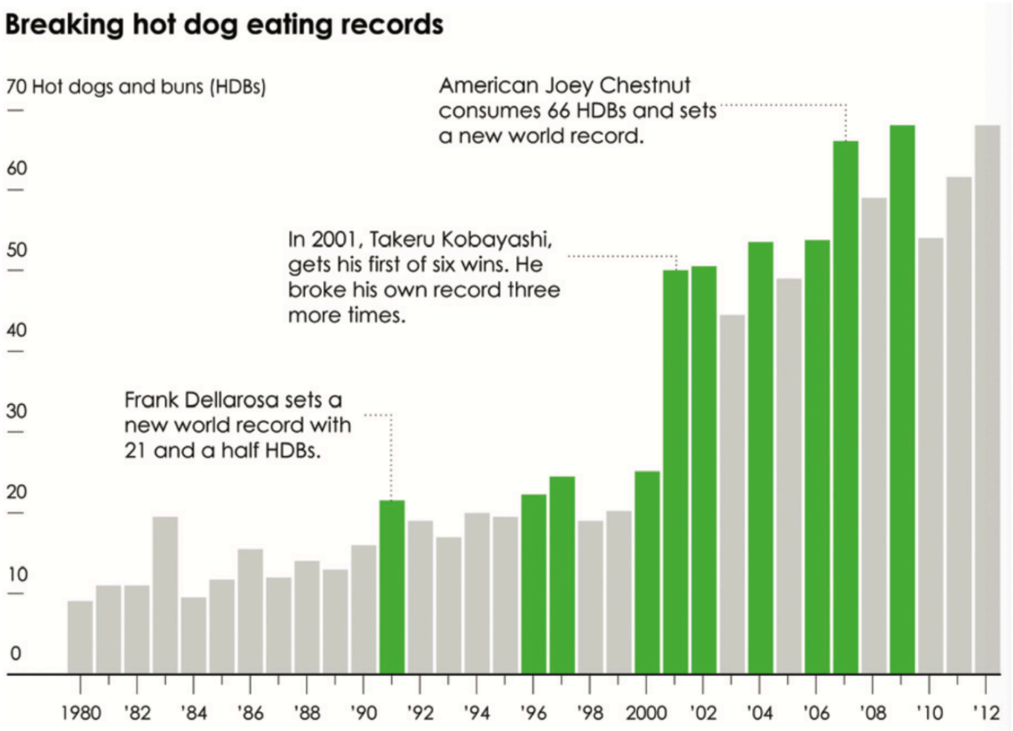


Kanizsa triangle



The visual system fills in missing boundaries to perceive complete objects.

Figure–ground determines what stands out



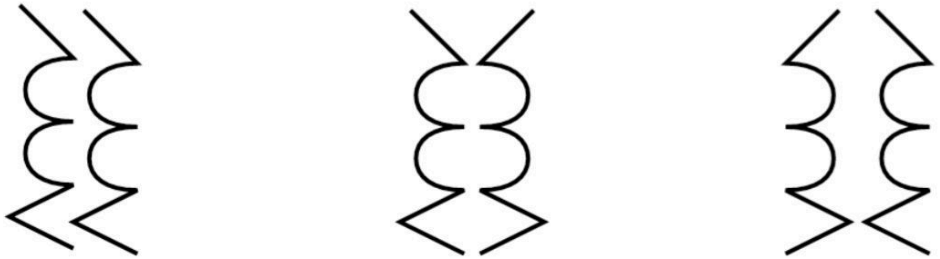
Clear figure–ground separation

Important elements stand out while supporting elements recede.

Ambiguous figure–ground

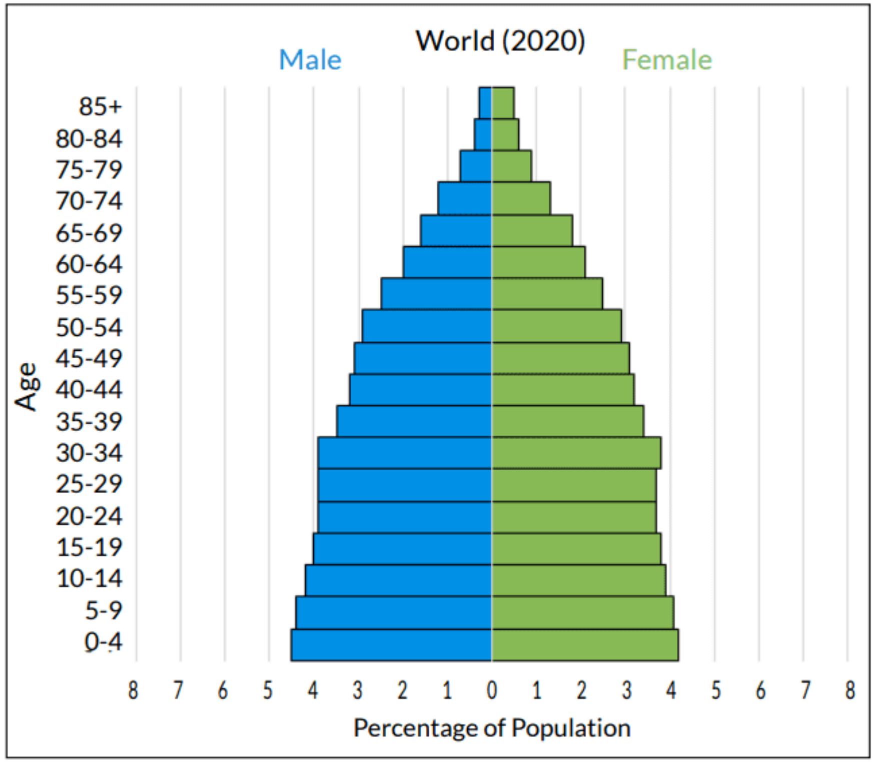
The viewer alternates between two interpretations.

Symmetry makes elements feel related



Symmetry creates a visual whole.

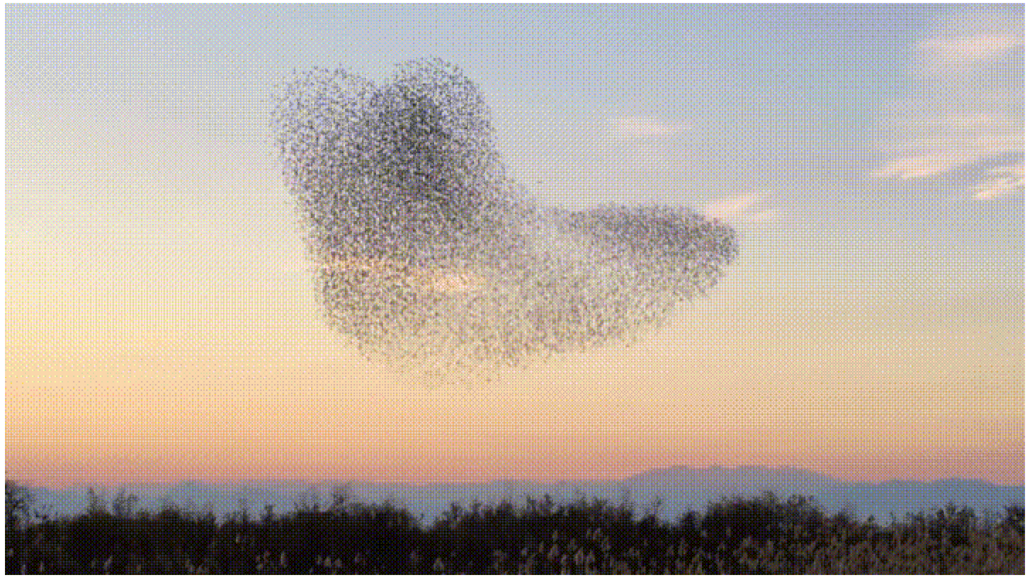
The centre and right pairs group more strongly than the parallel lines on the left.



Mirrored layouts support comparison.

Population pyramids use symmetry to compare two sides.

Common fate groups elements that move together



Elements moving in the same direction are perceived as a group.

Shared direction implies shared behavior.

Motion and orientation can reveal a group even when its elements are separated.

Gestalt laws: three organizing questions

01 · GROUPS

How elements form groups

Which elements are perceived as belonging together?

- **Proximity**
- **Similarity**
- **Connectedness**
- **Enclosure**

02 · OBJECTS

How the mind forms coherent objects

How does incomplete or complex visual evidence become an object?

Continuity
Follow smooth paths across fragments.

↓

Closure
Complete objects despite missing boundaries.

↓

Figure–Ground
Separate the object from its background.

03 · OTHER CUES

Additional grouping cues

What else makes elements feel related?

- **Symmetry**
- **Common Fate**

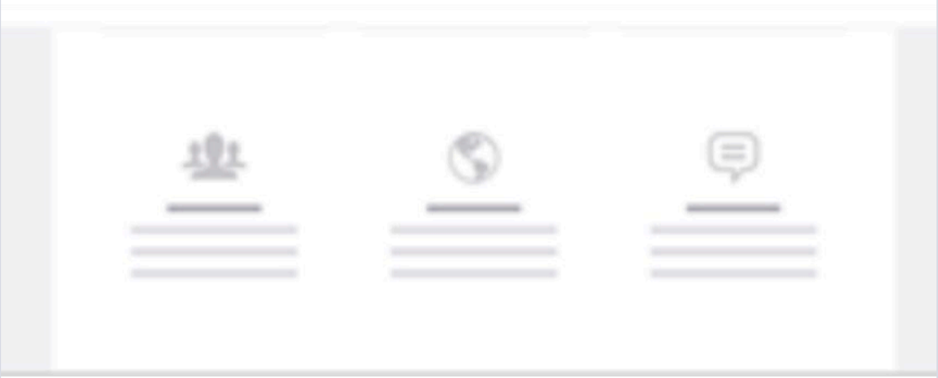
Which layout communicates six groups?

A

B

Vote first · Then justify with a Gestalt principle

Which layout treats three reports equally?



A

B

Discuss in pairs · Name the cue that creates hierarchy

Part 03

Turn principles into decisions

The goal is not to decorate—it is to control what viewers notice and infer.

From perception to design

What we learned	Design implication
Attention is selective	Make important information stand out.
Proximity / Similarity	Make related elements appear related.
Connectedness / Enclosure	Make relationships and groups explicit.
Continuity	Make paths easy to follow.
Figure–Ground	Keep important elements in the foreground.
Prior knowledge	Design for the viewer’s task and expectations.

Design the viewer’s perceptual path

01

Notice

What attracts attention first?

Selective attention
Bottom-up and top-down processing

02

Group

What elements appear to belong together?

Gestalt grouping principles

03

Follow

What relationships or paths are easy to trace?

Continuity
Connectedness

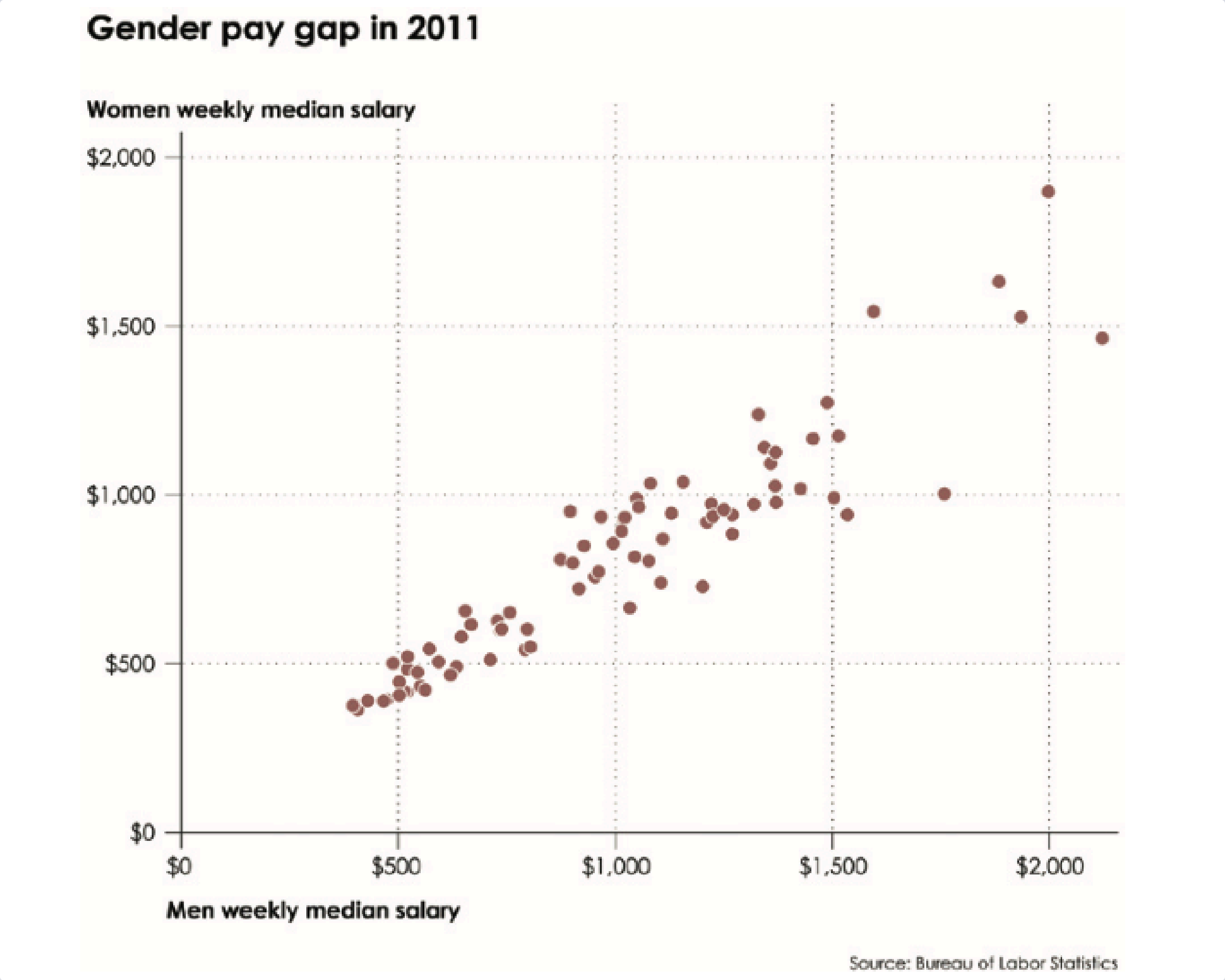
04

Interpret

Does the perceived structure support the intended meaning?

Prior knowledge
Visual cognition

Diagnose this visualization



1 · Notice

What do you notice first?

2 · Group

What appears grouped?

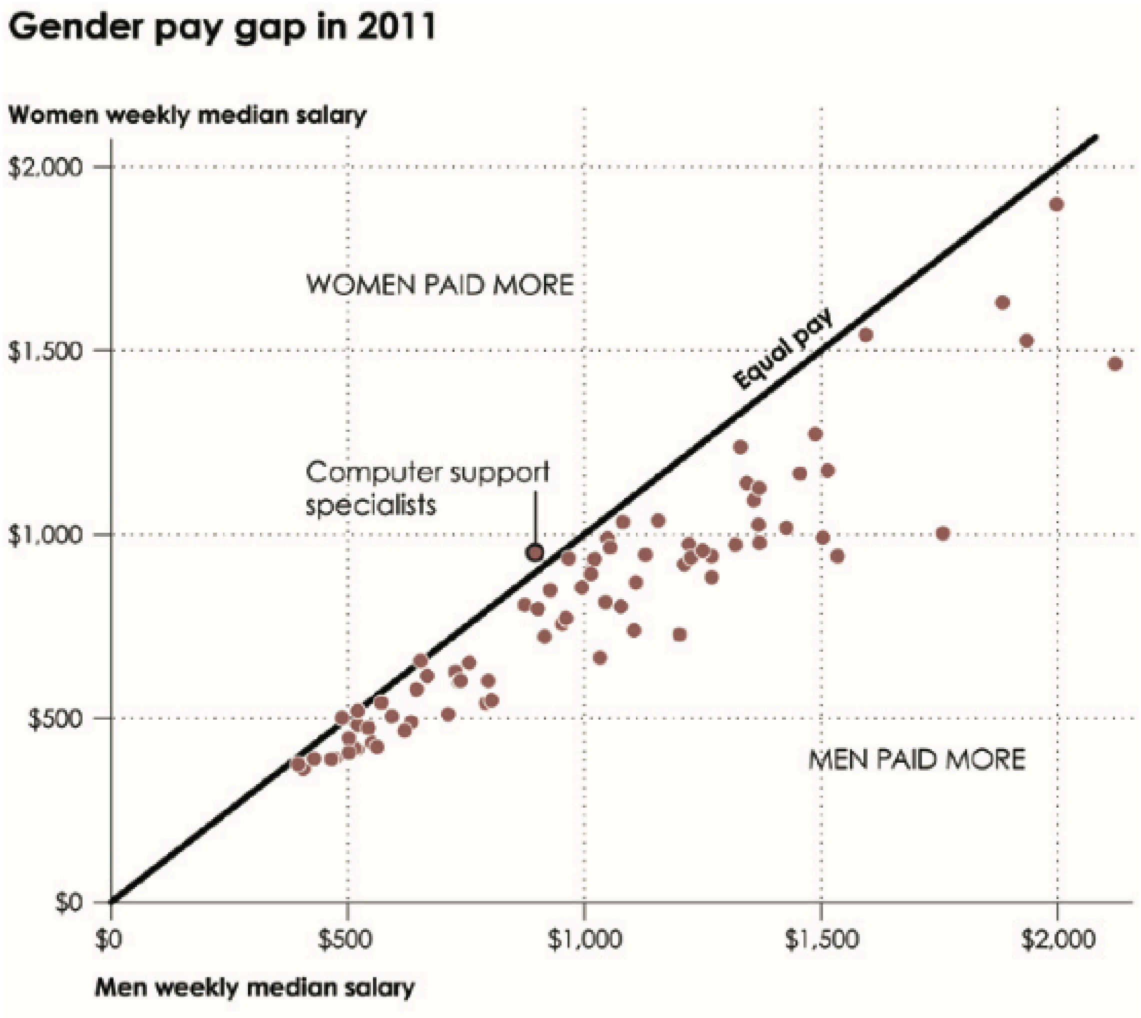
3 · Follow

What paths or relationships do you follow?

4 · Interpret

Does this support the intended interpretation?

Diagnose this visualization



1 · Notice

What do you notice first?

2 · Group

What appears grouped?

3 · Follow

What paths or relationships do you follow?

4 · Interpret

Does this support the intended interpretation?

Preview

How does data become visible?

The next two lectures move from the human visual system to the designer’s visual vocabulary.

Visual encoding maps data to marks and channels

Data attributes	→	Marks	→	Channels
category · quantity · time		points · lines · areas		position · color · size · shape

Today: why the viewer perceives patterns. Next: how the designer creates them.

Three ideas to carry forward

1. Seeing is **selective**, not passive.
2. Perception is shaped by both **visual evidence** and **prior knowledge**.
3. Layout creates meaning through **grouping, separation, and hierarchy**.